N5

National Qualifications Archives
RAPID-FIRE PAPER

Physics
Space, Waves, & Radiation

Duration: 58 minutes & 30 seconds

Total marks — 39

SECTION 1 — Space — 15 marks

Attempt ALL questions.

SECTION 2 — Waves — 10 marks

Attempt ALL questions.

SECTION 3 — Radiation — 14 marks

Attempt ALL questions.

Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. You should score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

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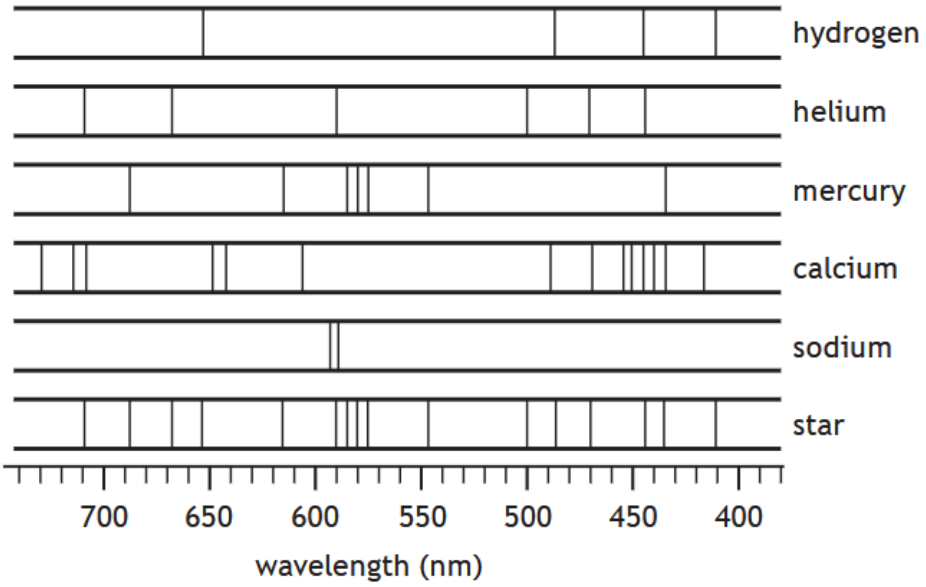
SECTION 1 — Space — 15 marks

Attempt ALL questions

4. Astronomers studying a distant star analyse the light from the star that reaches Earth.

The line spectrum from the star is shown, along with the line spectra of the elements hydrogen, helium, mercury, calcium, and sodium.



(a) Determine which of these elements are present in the star. **1**

(b) The star is 97 light-years from Earth.

(i) State what is meant by the term *light-year*. **1**

(ii) Calculate the distance, in metres, from the star to Earth. **3**

Space for working and answer

[Turn over

(c) Astronomers use satellite-based telescopes to collect information about objects in space.

(i) Suggest an advantage of using satellite-based telescopes such as the Hubble Space Telescope. **1**

(ii) State one **other** use of satellites. **1**

(b) In the future it is hoped that humans will be able to travel to Mars. One challenge of space travel to Mars is maintaining sufficient energy to operate life support systems.

(i) Suggest one solution to this challenge. **1**

(ii) State another challenge of space travel to Mars. **1**

12. The star Wolf 359 is at a distance of 7·8 light-years from Earth.

A radio signal from Wolf 359 is detected by a radio telescope on Earth.

(a) (i) State the speed of the radio waves. **1**

(ii) Calculate the distance, in metres, from Wolf 359 to Earth. **3**

Space for working and answer

(b) Another telescope is used to observe the same star in the visible part of the spectrum.

(i) State a suitable detector of visible light that may be used in this telescope. **1**

(ii) State whether the time taken for the visible light from the star to reach Earth is less than, equal to, or greater than the time taken for the radio waves from the star to reach Earth. **1**

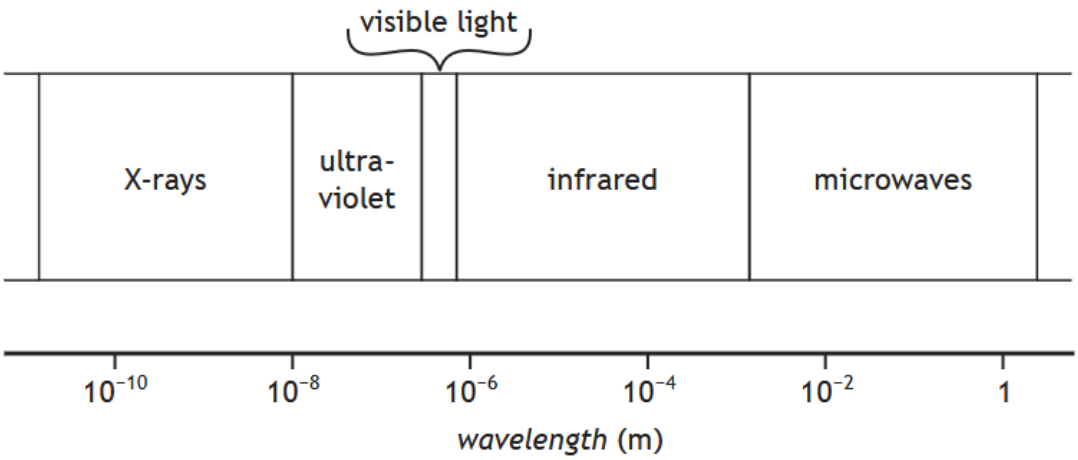
[END OF SECTION 1]

[TURN OVER AND ATTEMPT SECTION 2]

SECTION 2 – Waves – 10 marks

Attempt ALL questions

4. The diagram shows some parts of the electromagnetic spectrum in order of increasing wavelength.



(a) State a detector of infrared radiation. **1**

(b) State which radiation in the electromagnetic spectrum has a wavelength shorter than X-rays. **1**

(c) (i) An electromagnetic wave has a frequency of 1·2 GHz.

Show that the wavelength of this wave is 0·25 m. **2**

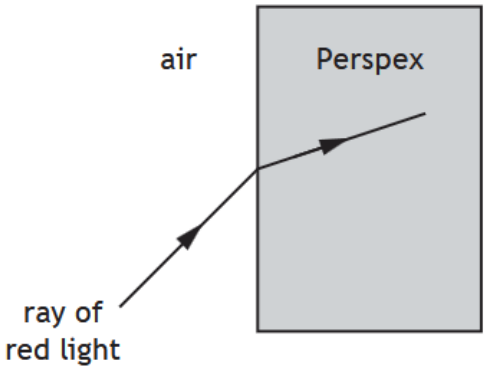
Space for working and answer

[Turn over

4. (c) (continued)

(ii) Identify the part of the spectrum that this wave belongs to. **1**

6. A student directs a ray of red light into a Perspex block to investigate refraction.



(a) On the diagram, draw and label:

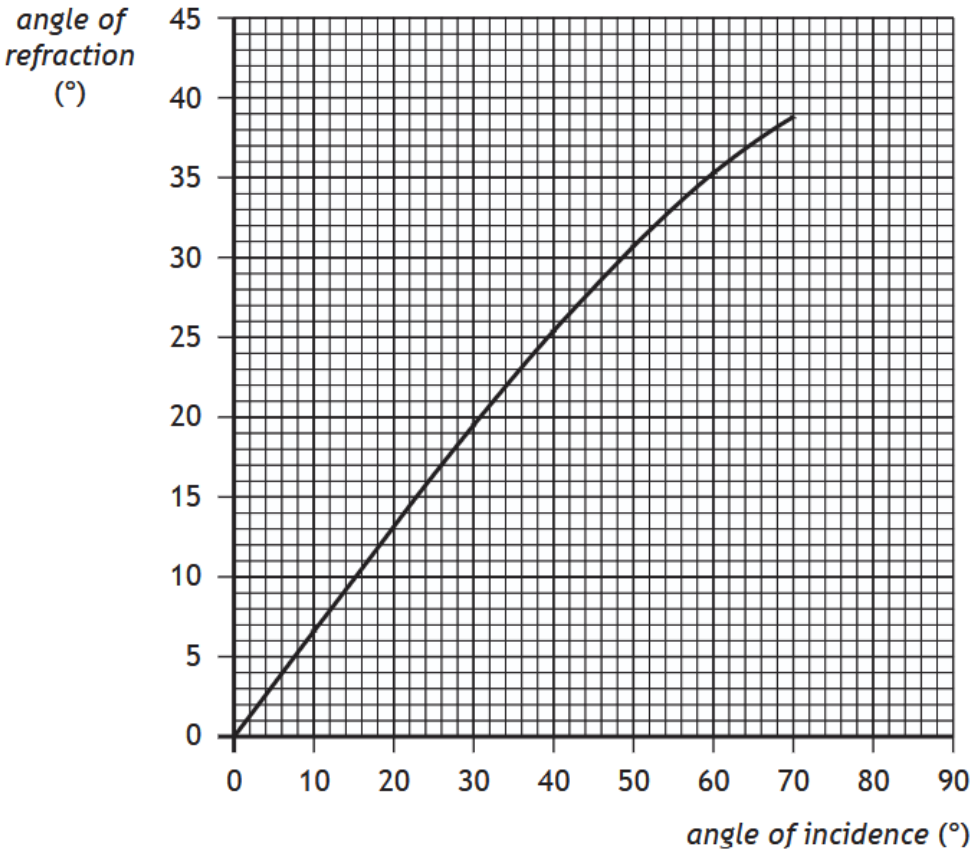
(i) the normal; **1**

(ii) the angle of incidence *i* and the angle of refraction *r*. **1**

(An additional diagram, if required, can be found on *Page 33*)

[Turn over

(b) The student varies the angle of incidence and measures the corresponding angles of refraction. The results are plotted on a graph.



(i) Determine the angle of refraction when the angle of incidence is 12°. **1**

(ii) Use the graph to predict the angle of refraction the student would obtain for an angle of incidence of 80°. **1**

(c) Suggest why it would be good practice for the student to repeat the investigation a further three or four times. **1**

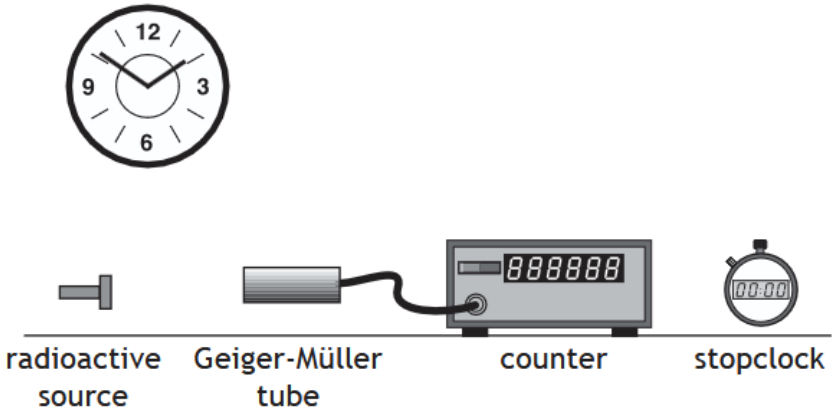
[END OF SECTION 2]

[TURN OVER AND START SECTION 3]

SECTION 3 – Radiation – 14 marks

Attempt ALL questions

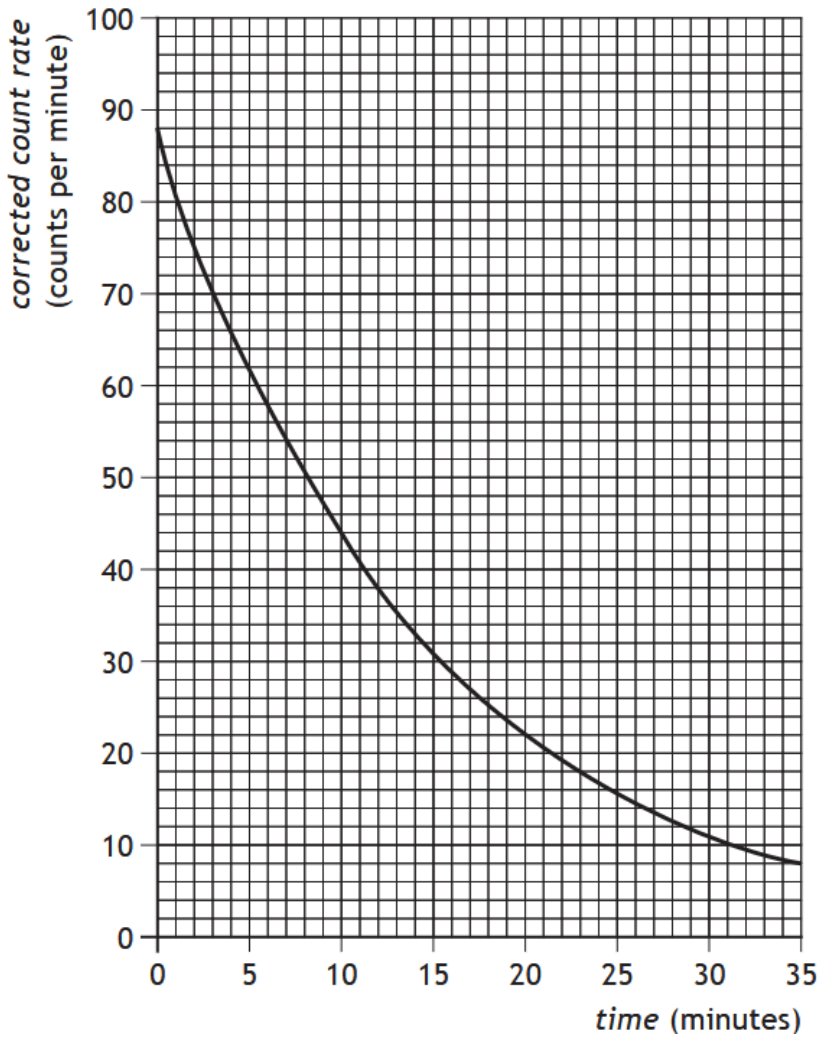
6. A technician carries out an experiment, using the apparatus shown, to determine the half-life of a radioactive source.



(a) State what is meant by the term *half-life*. **1**

[Turn over

(b) The technician displays the data obtained from the experiment in the graph below.



(i) Describe how the apparatus could be used to obtain the experimental data required to produce this graph. **3**

[Turn over

6. (b) (continued)

(ii) Use information from the graph to determine the half-life of the radioactive source. **1**

(iii) Determine the corrected count rate after 40 minutes. **2**

Space for working and answer

8. An airport worker passes suitcases through an X-ray machine.

(a) The worker has a mass of 80·0 kg and on a particular day absorbs 7·2 mJ of energy from the X-ray machine.

(i) Calculate the absorbed dose received by the worker. **3**

Space for working and answer

(ii) Calculate the equivalent dose received by the worker. **3**

Space for working and answer

[Turn over

8. (continued)

(b) X-rays can cause ionisation.

Explain what is meant by *ionisation*. **1**

[END OF SECTION 3]

[END OF QUESTION PAPER]